

Self-Supervised Learning and Multisource Heterogeneous Information Fusion Based Quality Anomaly Detection for Heavy-Plate Shape

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Abstract—Plate shape quality anomaly detection suffers from the problems of less labeled and multisource heterogeneous (MSH) data due to the complex production process with multi-process and multi-equipment. To address these problems, this paper proposes a self-supervised learning framework based on MSH contrastive learning (MSH-CL). In the proposed framework, an encoder is designed to extract the MSH information, which contains an MSH feature extract network (MSH-FEN) and an MSH feature fusion network (MSH-FFN). In MSH-FEN, a two-pathway CNN network is employed to extract the MSH feature. Then, a cross-MSH (CMSH) data fusion strategy based on the attention mechanism is designed to characterize the relationship between the MSH data in the MSH-FFN. The learning process consists of two phases, namely the self-supervised feature learning phase based on the contrastive learning (CL) and the supervised fine-tuning phase. The former phase is responsible for training the encoder with massive unlabeled data. Then, a classifier followed by the encoder is built with less labeled data in the supervised fine-tuning phase, which is applied to plate shape quality anomaly detection. Finally, the experiments are carried out on the real-world data set collected from a heavy-plate production process. The experimental results demonstrate the effectiveness and superiority of the proposed approach.

Note to Practitioners—This paper focuses on the problem of shape quality anomaly detection for the heavy-plate production process, which is crucial to steel-making. However, the most quality-relevant data-driven modeling approaches suffer from the problem of requiring massive labeled data. In addition, the data generated in the practice heavy-plate production process have the characteristics of multisource heterogeneous (MSH). Motivated by the challenge of the less labeled and MSH data modeling, this paper proposed a self-supervised feature learning approach to learn a feature encoder to model the MSH data. Then, fine-tuning is used to obtain the model to detect the plate shape quality. In the feature encoder, a cross-MSH (CMSH) data fusion strategy is proposed. Finally, the effectiveness of the proposed approach is

demonstrated by comparing it with other advanced algorithms. The proposed approach can be extended to other heavy-plate quality anomaly detection problems with less labeled and MSH data.

Index Terms—Heavy-plate, plate shape, multisource heterogeneous data, information fusion, self-supervised learning.

I. INTRODUCTION

HEAVY-PLATE has been widely applied in ships, chemicals, buildings, etc. In the past few decades, studies on the issues of the heavy-plate production process have received extensive attention, such as feedback control [1], operation optimization [2], and production scheduling [3], [4]. Recently, researchers have attached the importance of the problem of heavy-plate quality anomaly detection because of the strict requirements of the contract. Generally, the quality of heavy-plate includes five major indicators, namely shape, size, performance, surface, and internal quality. Among them, the plate shape is one of the most important and complex quality indicators and is closely related to the production process. Currently, the plate shape quality is determined by manually evaluating the final quality control (FQC) image obtained from a shape meter at the end of the production process, resulting in an unavoidable time delay. Therefore, detecting the plate shape quality conveniently and accurately is of great significance to heavy-plate production.

The plate shape quality is difficult to detect, because the production process consists of multi-equipment, and has the characteristics of small batch and multi-variety. Therefore, it is hard to establish an accurate mechanism model. In modern industries, a large amount of data has been collected from the actual production process, and the quality of products generally can be reflected by process variables. The data-driven methods are more flexible than mechanism models [5]. Therefore, these methods have been widely used to address the issues in the industrial field, such as fault diagnosis [6], [7], process monitoring [8] and soft sensor [9], [10], which are also available for quality prediction [11], [12], [13], [14]. For example, Lu and Ding [15] proposed a novel stochastic configuration network with iterative learning using privileged information for the plate shape quality detection based on the process data.

The aforementioned methods depend decisively on massive labeled data, which require extensive manual annotation. How-

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ever, it is always difficult to obtain the labels of heavy-plate quality, since only a part of the plates will be detected by the shape meter. Thus, the production process data present the less labeled characteristic. In addition, the multi-equipment involved in the heavy-plate production process leads to the diverse data sources. Moreover, the size of the steel plates will change with the slabs, and the sampling frequency and the running speed of roller also vary with different production processes. As a result, the data structures collected from the actual production process of steel plates exhibit multisource heterogeneous (MSH) characteristic. Therefore, the above imperfect attributes of the process data pose the following challenges for detecting the heavy-plate quality:

- 1) **Less labeled data.** Most data-driven methods for industrial process modeling require a lot of labeled data, which is often hard to acquire. Therefore, an intricate problem is how to use the massive unlabeled process data to assist in building a data-driven model to detect the heavy-plate shape.
- 2) **MSH process data.** It is challenging to extract the features of the data with MSH characteristic. Thus, another difficult task is designing an effective feature extraction and fusion approach for MSH data.

Recently, many researchers have paid attention to the above challenges. For example, Bao et al. [16] proposed a co-training partial least squares (PLS) for developing semi-supervised soft sensors. Two PLS models are trained on two sufficient and redundant views using the iterative learning procedure for augmenting the training set, respectively. On this basis, two new regressors are constructed for soft sensing. However, when the labeled data is insufficient, the correctness of the labels obtained by such an iterative method cannot be guaranteed, which will degrade the performance of the learner model. With the development of pre-training trick and transfer learning, the fine-tuning strategy is another candidate for solving the problem of less labeled data. For example, Yuan et al. [17] presented a stacked autoencoder based quality prediction model with fine-tuning strategy. However, they did not consider the MSH problem in real-world industrial data. Because in the pre-training stage, the encoder must be trained by calculating the reconstruction error between the decoded and original data. Thus, it is difficult to establish an encoder to model MSH data. More detailed methods of modeling the less labeled industrial process data can be found in [18].

The MSH process data is present in many modern industries, which also received much concerns [19], [20], [21], [22], [23]. To model the MSH data, for example, the heterogeneous data are transformed into linguistic 2-tuples, but the high-dimensional data obtained in the sensor is not taken into account [24]. Yuan et al. [20] develop a comprehensive decision-making framework with multi-source heterogeneous data for selecting the most appropriate distributed energy system composition schemes. Pan and Zhang [21] present a predictive method to accurately estimate building energy consumption by learning MSH data collected from buildings. Furthermore, Bu et al. [25] constructed a method for operating performance assessment based on MSH data, the multi-source

data features are extracted by different models. Besides, Zhou et al. [26] use the deep learning method to extract the features of multisource information and directly concatenate the features of different branches as the final representation. However, they did not consider the correlation between MSH data.

Contrastive learning (CL) has become a popular self-supervised representation learning framework in recent years and achieved outstanding performance in multiple computer vision tasks [27], [28], [29], [30]. In the CL framework, a feature encoder is learned using data augmentation and contrastive loss to obtain the latent codes of the samples. Then, the encoder is used in downstream tasks. The contrastive loss expects to repulse negative sample pairs while attracting the positive pairs. In [28], a sample CL framework, named SimCLR, is proposed, which combines the multiple data augmentation methods and a nonlinear projector to obtain the feature representation. The above studies indicate that such a CL learning paradigm can obtain more reasonable feature representations. It is essential to design corresponding data augmentation according to the characteristics of the actual problem, which will influence the model preferences [29]. In recent studies, the CL has been used to deal with industrial data modeling issues. For example, contrastive loss is used as a regularization term to train a soft sensing model [31]. As far as we know, CL is not adopted to solve both the problems of less labeled and MSH data for industrial data-driven modeling.

To this end, this paper proposes a novel multisource heterogeneous contrastive learning (MSH-CL) framework, which is a typical self-supervised representation learning paradigm, to detect the abnormal quality of heavy-plate shape. The contributions of this paper are summarized as follows:

- 1) The MSH-CL framework is designed to handle the issues of less labeled and MSH data by incorporating CL learning, and multiple components are developed.
- 2) An MSH feature encoder (MSHFE) that consists of an MSH feature extract network (MSH-FEN) and an MSH feature fusion network (MSH-FFN) is developed to extract the MSH information.
- 3) In the MSH-FEN, a two-pathway network is developed to extract the feature of MSH data. After that, in the MSH-FFN, an attention mechanism based cross-MSH (CMSH) data fusion strategy is presented to characterize the relationship between the MSH data.
- 4) The effectiveness of the proposed approach is verified on the real-world data collected from the heavy-plate production process.

The rest of this article is organized as follows. First, the process of heavy-plate production is described in Section II. Then, the framework of quality anomaly detection for heavy-plate shape is presented in Section III. In Section IV, the experimental results are presented and discussed. Finally, the conclusion is given in Section V.

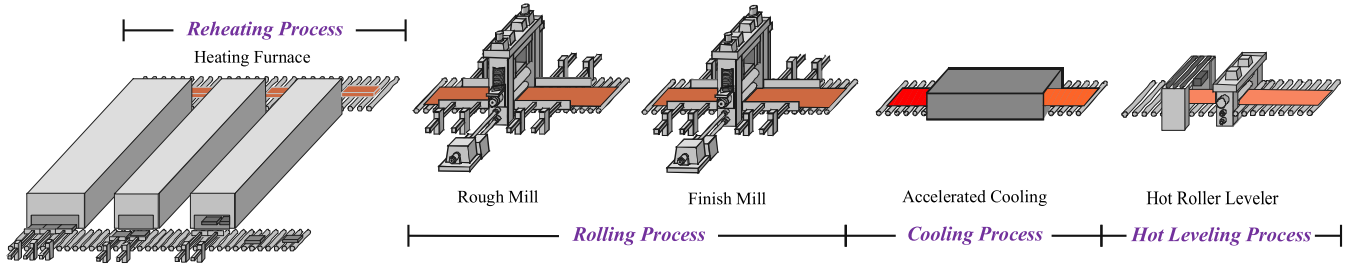


Fig. 1. The primary production process of heavy-plate.

II. PROBLEM STATEMENT

The shape quality is highly relevant to the heavy-plate production process [32]. In this section, the primary process of heavy-plate production is first briefly described. Then, the relationship between the process and the plate shape, and the characteristics of the production process data are introduced.

A. Description of Heavy-Plate Production Process

The production of heavy-plate is relatively complicated, which includes multiple processes. A practical production process of heavy-plate in Eastern China is shown in Fig. 1, which mainly consists of reheating, rolling, cooling and hot leveling process.

The reheating process contains preheating section, heating section I, heating section II and soaking section in the heating furnace. The steel slab enters the heating furnace and is first heated to the target rolling temperature in the reheating process. The following process behind the reheating process is the rolling process, which contains a rough mill and a finish mill, and processes the heated slab through the three stages of forming, widening and finishing rolling. It is rolled repeatedly to produce a steel plate that meets the required size and shape. Subsequently, the cooling process follows the rolling process, which contains an accelerated cooling device that can reduce the temperature of the plate rapidly and accurately. The steel plate is sent into the cooling process, and the plate temperature is dropped rapidly to satisfy the designed finish cooling temperature and the requirements of low-temperature toughness, strength, and welding performance. The hot leveling process that contains a hot roller leveler is set behind the cooling process, and the cooled plate is straightened to a certain extent in this process. Therefore, the shape quality of the heavy-plate is mainly related to the processes mentioned above.

B. Description of the Heavy-Plate Shape Quality and Data Characteristics

The shape quality of a heavy-plate can be defined by the flatness, rectangularity and plate crown. Without loss of generality, the abnormal shape quality is essentially caused by the uneven distribution of the internal stress of the steel plate. The significant factors influencing the shape in the production process are the temperature and the rolling force, such as the uniformity of the temperature of the slab, the uneven internal stress caused by the rolling mill, the plate temperature uniformity before and after cooling, and so on.

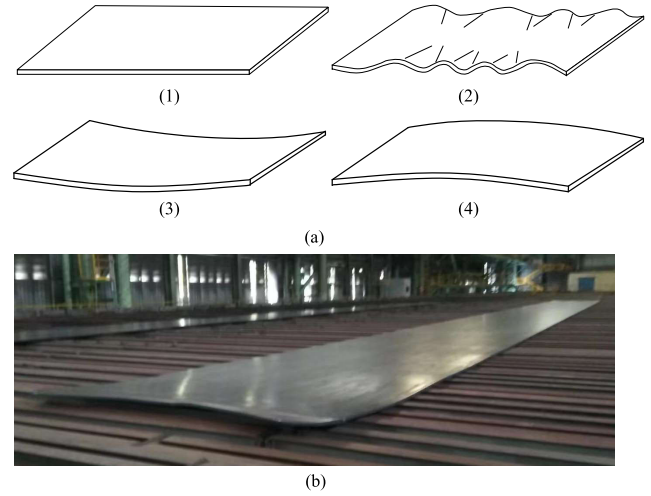


Fig. 2. The common shape quality of the heavy-plate. (a) The schematic diagram of plate shape qualities. (1) The normal shape quality. (2) The abnormal shape quality of the wave. (3) and (4) present the abnormal quality of the warpage. (b) A real-world case of the abnormal heavy-plate shape.

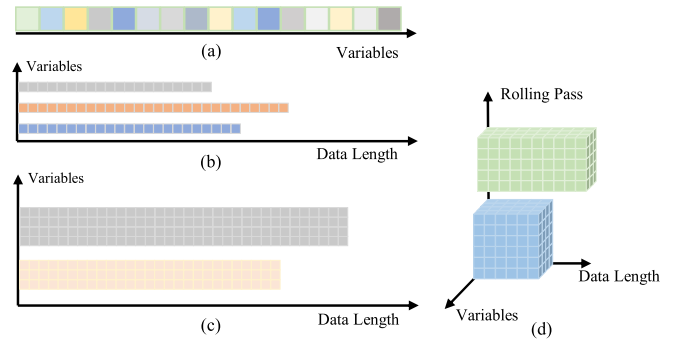


Fig. 3. The schematic diagram of the MSH data collected from the heavy-plate production process. (a) The real number data. (b) The one-dimensional data. (c) The two-dimensional data. (d) The three-dimensional data.

The common qualities of the heavy-plate shape are shown in Fig. 2. However, acquiring labels of heavy-plate shape quality is commonly expensive, which is one of the challenging problems.

The essential to building a heavy-plate shape quality anomaly detection model is to establish the corresponding relationship between the process variables and the final plate shape quality. Therefore, it is necessary to analyze the production process and process variables of the heavy-plate and select appropriate process variables.

Since heavy-plate production involves multiple processes and equipments, the data structure is various and complicated, which presents the MSH characteristic. On the one hand, the data structures of intra-sample x can be simply divided into four categories: the real number data $x_1, x_2, \dots, x_n$, one-dimensional data $y_1, y_2, \dots, y_a$, two-dimensional data $z_1, z_2, \dots, z_b$, and three-dimensional data $v_1, v_2, \dots, v_c$, as shown in Fig. 3. For example, the statistics of the temperature of plate surface, roller table, environment, and other indicators are the real number data; the temperature of start cooling, finish cooling, and other indicators acquired by sensors are the one-dimensional data; the temperature of finish cooling, heating furnace and other indicators acquired by sensors are shown as the two-dimensional data; the rolling force, roll bending force, torque and other indicators are present as the three-dimensional data. Thus, the real number data is commonly the key process variable, and the difference in the data dimension is caused by the different types of sensors and the manufacturing process. On the other hand, the data length of inter-samples is also different due to the distinct plate lengths and rolling speeds. Therefore, another considerable challenge is how to deal with the MSH process data.

Meanwhile, by analyzing the influence of each process on the plate shape quality and the data characteristics of the relevant variables, the process variables related to the plate shape are screened out. Then, the corresponding anomaly detection model of the plate shape quality can be built using the process data.

III. METHODOLOGY

The overall structure of the proposed MSH-CL based self-supervised representation learning approach for the shape quality anomaly detection of heavy-plate is described in Fig. 4. First, the data is collected from the heavy-plate production process and the shape quality-relevant variables are selected. Then, the selected data are cleaned and preprocessed before training process. After that, the proposed MSH-CL is adopted to build the quality anomaly detection model for the heavy-plate shape. It contains two learning phases, namely the CL phase and the fine-tuning phase, and one quality anomaly detection phase. The CL phase aims to learn an MSHFE with massive unlabeled data. Then, the trained MSHFE is used to build the detection model with the less labeled data in the fine-tuning phase. Finally, the trained model is implemented to detect the shape quality. This section mainly introduces the data cleaning and preprocessing method, the MSH-CL framework that includes an MSH-FEN and an MSH-FFN, and the implementation process for quality anomaly detection of plate shape.

A. Multisource Heterogeneous Data Cleaning and Preprocessing

The process data collected from the real-world heavy-plate production environment have the characteristic of MSH. Thus, the data should be cleaned and preprocessed first.

1) *Data Cleaning*: The process data collected from real-world industries should be cleaned first due to the complex production environment, and the sample containing the null value will be eliminated.

2) *Data Preprocessing*: After data cleaning, the MSH data examples should be preprocessed before modeling. As the analysis in Section II-B, the MSH data in heavy-plate production can be divided into two parts. First, the real number variables $x_1, x_2, \dots, x_n$ are set as a vector $\mathbf{x}_{1d} = [x_1, x_2, \dots, x_n]$, where $\mathbf{x}_{1d} \in \mathcal{R}^{1 \times n}$, n is the number of variables. The linear interpolation is employed as the preprocessing approach to deal with the remaining variables $y_1, y_2, \dots, y_a, z_1, z_2, \dots, z_b, v_1, v_2, \dots, v_c$. The examples of preprocessed data after interpolation are shown in Fig. 5. The $y_1, y_2, \dots, y_a$ are preprocessed to $y'_1, y'_2, \dots, y'_a$, where is the $1 \times m$ vector, m is the data length after preprocessing. The $z_1, z_2, \dots, z_b, v_1, v_2, \dots, v_c$ are flattened and then linearly interpolated to a $1 \times m$ vector, where the data after preprocessing are $z'_1, z'_2, \dots, z'_b, v'_1, v'_2, \dots, v'_c$. Then, $y'_1, y'_2, \dots, y'_a, z'_1, z'_2, \dots, z'_b, v'_1, v'_2, \dots, v'_c$ are concatenated as a matrix $\mathbf{X}_{2d} = [y'_1, y'_2, \dots, y'_a, z'_1, z'_2, \dots, z'_b, v'_1, v'_2, \dots, v'_c]^\top$, where $\mathbf{X}_{2d} \in \mathcal{R}^{v \times m}$, $v = a + b + c$ is the number of variables.

The preprocessing approach adopted in this paper makes the MSH data convenient to deal with and extract the feature. Meanwhile, an MSH data example x is divided into two parts, $\mathbf{x}_{1d}$ and $\mathbf{X}_{2d}$, which can maintain the MSH characteristics of the original data to a certain extent.

B. Multisource Heterogeneous Contrastive Learning

The schematic diagram of the MSH-CL framework is shown in the right part of Fig. 4. In order to detect shape quality with less labeled data, the proposed approach consists of two learning phases: *CL* and *fine-tuning*, and a detecting phase: *quality anomaly detection*.

1) *CL Phase*: This phase is proposed to train a feature encoder $f_\theta(\cdot)$ with massive unlabeled data to latent code the MSH data. Fig. 6 shows the detailed architecture of CL in this paper, which is a Siamese network with shared weights θ . The two branches are identical, and each branch is comprised of an MSHFE $f_\theta(\cdot)$ and a nonlinear projector unit $g_\theta(\cdot)$. The added projector $g_\theta(\cdot)$ is a multilayer perceptron (MLP), which can improve the feature representation performance of the encoder $f_\theta(\cdot)$ [28].

The data augmentation $Aug(\cdot)$ is critical in CL and will affect the discriminative ability of quality features mined from process data. A sample x is augmented to produce two difference views x_{v1}, x_{v2} , which create similarity relations for the contrastive loss and define invariants that the model should learn. Furthermore, data augmentation for CL should not change the semantic information of the sample x . However, the augmentation methods for process data usually add Gaussian noise to the raw data, which may significantly affect the data distribution. Thus, data augmentation methods should be appropriately designed for CL (See Section IV-D1).

In the training phase of CL, firstly, a minibatch of N data examples is randomly sampled and augmented pair-wise,

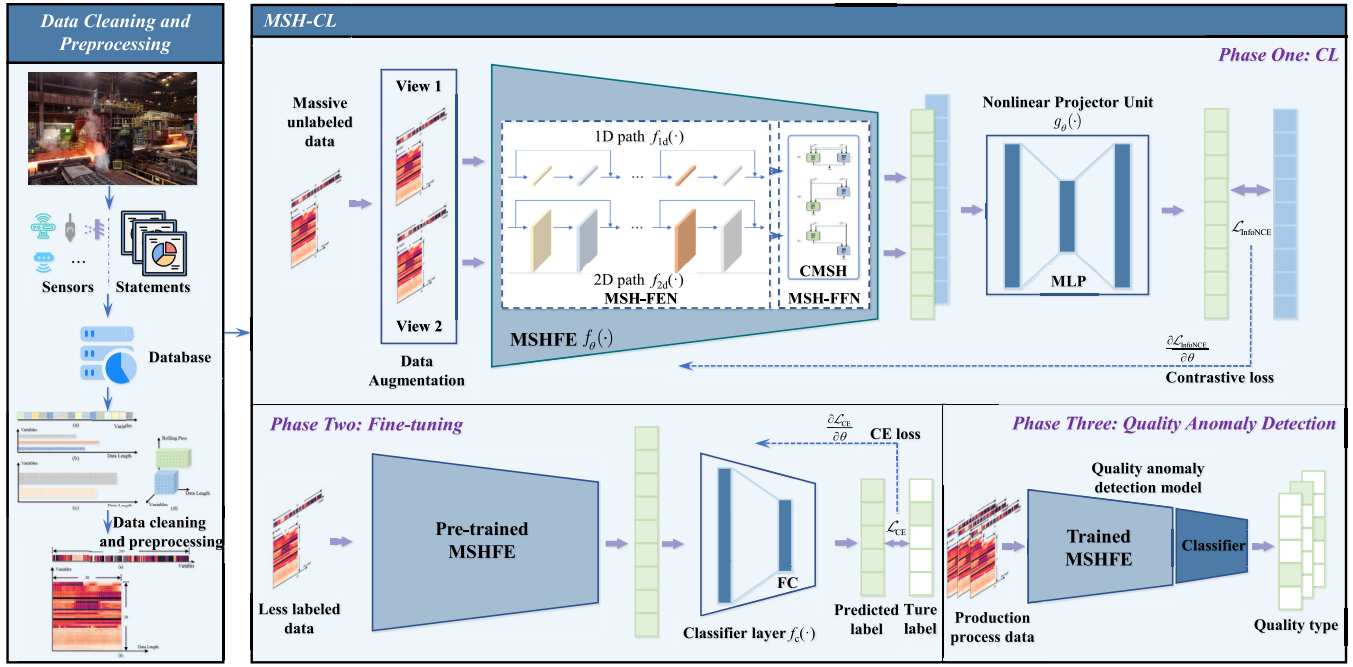


Fig. 4. The structure of the MSH-CL for quality anomaly detection.

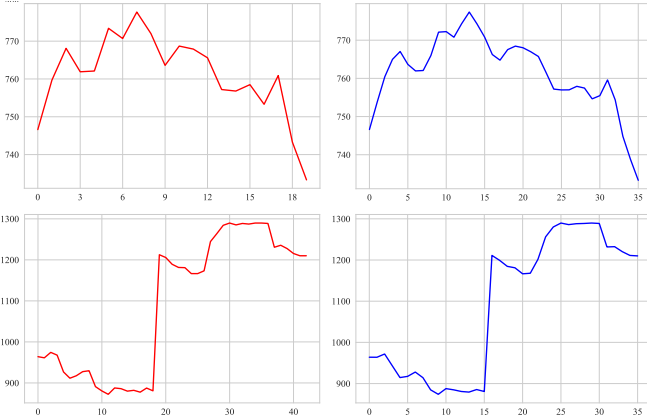


Fig. 5. The examples of MSH data after linear interpolation. The left column is the original data and the right column is the data after linear interpolation.

which results in $2N$ data examples. The two views x_{v1}, x_{v2} of the same data example are treated as a positive example pair, and the remaining $2(N-1)$ samples within the minibatch are viewed as negative examples. The negative examples in CL aim at preventing model collapse during the training process.

Subsequently, the augmented minibatch is coded by the feature encoder $f_\theta(\cdot)$ to produce the latent code h , and then the nonlinear projector $g_\theta(\cdot)$ is used to map h to feature z , which can be formalized as follow:

$$h = f_\theta(x) \quad (1)$$

$$z = g_\theta(h). \quad (2)$$

Then, z is used to calculate the contrastive loss. z_i and z_i^+ are the positive pair of the i th example. The negative example features within the minibatch are denoted as $z_i^- = \{z_j | j \neq i\}$. The cosine similarity $\text{sim}(u, v) = u^\top v / \|u\| \|v\|$ is adopted to

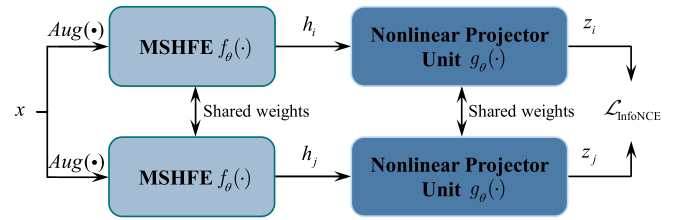


Fig. 6. The schematic diagram of pre-training phase in MSH-CL.

measure the similarity between each sample by the dot product of ℓ_2 normalized u and v . The typical contrastive loss function, named InfoNCE loss, is defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{InfoNCE}} &= \sum_{i=1}^N -\log \frac{\text{sim}_{\text{pos}}}{\text{sim}_{\text{pos}} + \text{sim}_{\text{neg}}} \\ \text{sim}_{\text{pos}} &= \exp(\text{sim}(z_i, z_i^+)/\tau) \\ \text{sim}_{\text{neg}} &= \sum_{z_j \in z_i^-} \exp(\text{sim}(z_i, z_j)/\tau) \end{aligned} \quad (3)$$

where τ denotes the temperature hyper-parameter. The InfoNCE loss can maximize the similarities between each positive pair and simultaneously minimize that between negative pairs. The weights θ of each branch are updated by the optimizer and backpropagation of the contrastive loss $\mathcal{L}_{\text{InfoNCE}}$. Therefore, the encoder $f_\theta(\cdot)$ and latent code h can be obtained by minimizing the loss using unlabeled data.

2) *Fine-Tuning Phase*: The trained encoder $f_\theta(\cdot)$ in the CL phase is taken out and follows a fully connected network as the classifier $f_c(\cdot)$ to establish the plate shape anomaly detection model. Subsequently, the model is fine-tuned by the less labeled data to realize the capacity of quality anomaly

detection. The cross entropy (CE) loss $\mathcal{L}_{CE} = -\sum_{i=1}^t p_i \log q_i$ is used in this training phase, where t denotes the total number of classes, p_i and q_i denote the true label and predicted probability output of i -th class, respectively.

3) *Quality Anomaly Detection Phase*: Finally, the trained model is deployed to detect the plate shape quality.

C. Multisource Heterogeneous Information Feature Encoder

In this part, the encoder $f_\theta(\cdot)$ is designed to draw the MSH information, which consists of two parts: the *MSH feature extract network* and the *MSH feature fusion network*.

1) *MSH Feature Extract Network*: The MSH data include multiple dimensions. Therefore, the two pathways feature extract network is proposed, which is formed by a 1D path and a 2D path to deal with the $\mathbf{x}_{1d}$ and $\mathbf{X}_{2d}$, respectively.

The convolutional neural networks (CNNs) are employed in this part as the feature extract network. The deep residual networks (ResNets) are an advanced CNNs architecture and have been used as the backbone to extract image features in various computer vision tasks [33]. ResNets contain two kinds of block, i.e., the residual building block and the bottleneck building block, and can be extended to different depths. Thus, the ResNet is a preferred candidate because of its flexibility and capability.

The 1D convolution is used in the 1D path to acquire the feature of $\mathbf{x}_{1d}$. Thus, a 1D vision of ResNet is used as the encoder $f_{1d}(\cdot)$. The naive ResNets are directly used as the encoder $f_{2d}(\cdot)$ in the 2D path to extract the feature of $\mathbf{X}_{2d}$. The two pathways of MSH-FEN can be formulated as follow:

$$h_{1d} = f_{1d}(\mathbf{x}_{1d}) \quad (4)$$

$$h_{2d} = f_{2d}(\mathbf{X}_{2d}). \quad (5)$$

where h_{1d} and h_{2d} denote the feature of $\mathbf{x}_{1d}$ and $\mathbf{X}_{2d}$, respectively. Then, h_{1d} and h_{2d} are used to obtain the latent code of the MSH data.

2) *MSH Feature Fusion Network*: In this part, a fusion network is designed to combine the two pathways network in the MSH-FEN.

The common fusion strategies to obtain the latent code h of an MSH data sample x are to add or concatenate the feature of each data structure directly. Such fusion approaches have often ignored the inner correlation between MSH data and cannot find out a better feature representation. Thus, the fusion strategy of MSH data should take advantage of the correlation between h_{1d} and h_{2d} .

The attention mechanism is an effective method in deep learning and is widely used in various tasks of computer vision and natural language processing. It is presented to capture the most relevant parts of visual or textual inputs like a human being [34], [35]. The attention mechanism improves the model performance by focusing on the valuable parts. Therefore, inspired by the superiority of the attention mechanism, an attention-based feature fusion network is proposed to bridge the feature h_{1d} and h_{2d} of MSH data.

Based on the attention mechanism, the cross-MSH (CMSH) data interaction approach is introduced in MSH-FFN to construct a high-level representation of the MSH data sample

x and improve the quality of latent code h . The CMSH consists of two attention modules, i.e., the low dimensional information-guided attention (LIGA) module and the high dimensional information-guided attention (HIGA) module, which are shown in Fig. 7. In these two modules, the underlying relationship between different data structures is captured to enhance the feature representation h_{1d} and h_{2d} , respectively.

In the LIGA, the information of h_{1d} is utilized to enhance the feature representation h_{2d} . First, for h_{1d} and h_{2d} , the corresponding query Q_{1d} , key K_{2d} and value V_{2d} can be generated by

$$Q_{1d} = W_{Q_{1d}} h_{1d}, \quad K_{2d} = W_{K_{2d}} h_{2d}, \quad V_{2d} = W_{V_{2d}} h_{2d} \quad (6)$$

where $W_{Q_{1d}}$, $W_{K_{2d}}$ and $W_{V_{2d}}$ are matrices with appropriate dimensions. Thereafter, Q_{1d} and K_{2d} are used to calculate the attention map S_{LIGA} of LIGA as

$$S_{LIGA} = \text{softmax}(Q_{1d} K_{2d}^\top). \quad (7)$$

Finally, the output V'_{2d} of LIGA is the weighted sum with the value V_{2d} as

$$V'_{2d} = S_{LIGA} V_{2d}. \quad (8)$$

Contrast to LIGA, h_{1d} is enhanced by h_{2d} in the HIGA, of which query Q_{2d} , key K_{1d} and value V_{1d} can be formulated as

$$Q_{2d} = W_{Q_{2d}} h_{2d}, \quad K_{1d} = W_{K_{1d}} h_{1d}, \quad V_{1d} = W_{V_{1d}} h_{1d} \quad (9)$$

where $W_{(\cdot)}$ denotes the weight matrix of $(\cdot)$ as in (6). Similarly, the attention map S_{HIGA} and the output V'_{1d} of HIGA are

$$S_{HIGA} = \text{softmax}(Q_{2d} K_{1d}^\top) \quad (10)$$

$$V'_{1d} = S_{HIGA} V_{1d}. \quad (11)$$

Then, in order to better develop the CMSH, three strategies are proposed to integrate the LIGA and HIGA for constructing the MSH-FFN, namely *parallel*, *first HIGA then LIGA*, and *first LIGA then HIGA*. The CMSH is shown in Fig. 8. Different strategies can make the CMSH adapt to diversified tasks.

D. Implementation Process for Quality Anomaly Detection

The MSH-CL for quality anomaly detection can be summarized in the following four steps:

- 1) The collected process data is cleaned and preprocessed. Then, the data is divided into labeled data and unlabeled data.
- 2) In the training phase of CL, the unlabeled data is used to train the MSHFE $f_\theta(\cdot)$. First, the unlabeled data is generated in different views through data augmentation. Then, the generated data is input into the Siamese network, and the InfoNCE loss is selected as the cost function. Finally, $f_\theta(\cdot)$ is trained by utilizing an optimizer to update the weight θ via backpropagation. The CL phase is shown in Algorithm 1.
- 3) In the fine-tuning phase, the trained MSHFE $f_\theta(\cdot)$ is taken out and followed by a classifier $f_c(\cdot)$, and then the model is fine-tuned by the labeled data.

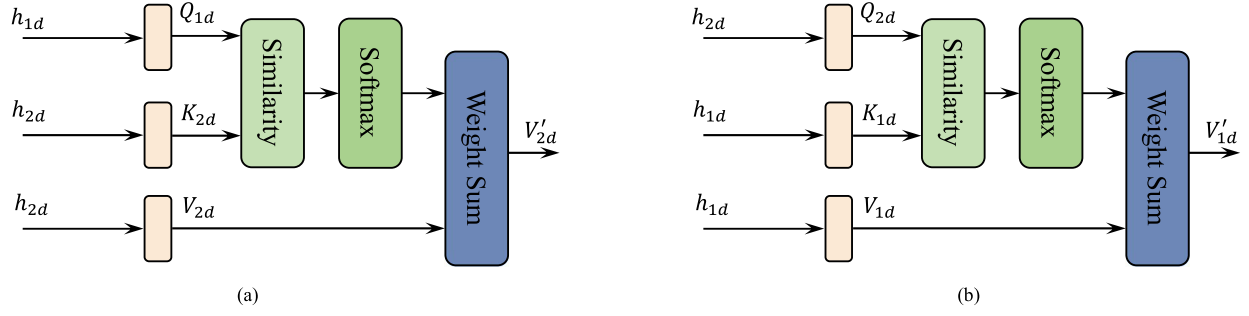


Fig. 7. The schematic diagram of LIGA and HIGA. (a) LIGA. (b) HIGA.

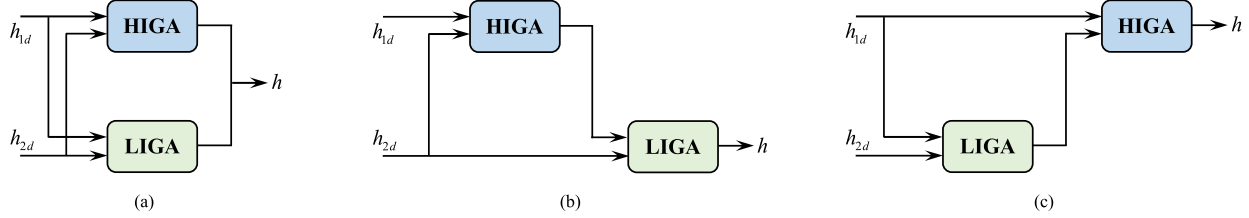


Fig. 8. The schematic diagram of CSMH fusion strategy. (a) Parallel. (b) First HIGA then LIGA. (c) First LIGA then HIGA.

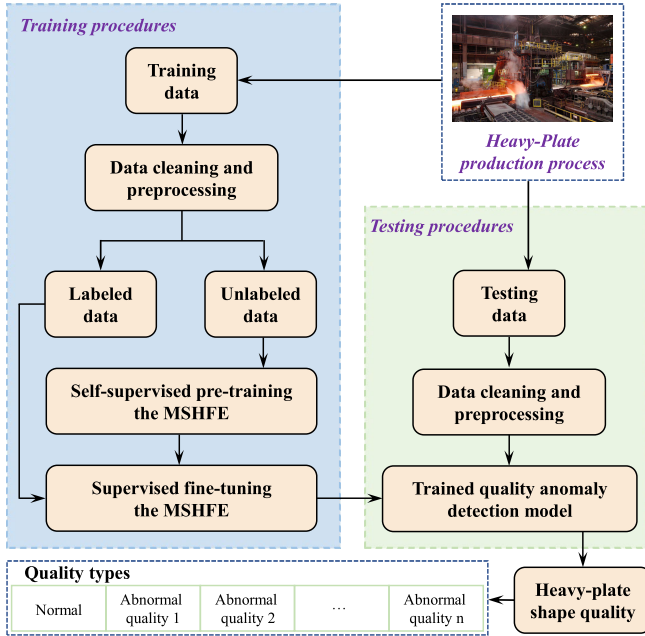


Fig. 9. The flow chart of MSH-CL based quality anomaly detection.

- 4) When the production of a new heavy-plate is finished, the obtained model is used to detect the shape quality of this plate.

The flow chart of MSH-CL-based quality anomaly detection can be seen in Fig. 9.

IV. EXPERIMENTS ON ACTUAL PRODUCTION DATA

In this section, experiments are carried out to demonstrate the effectiveness of the proposed MSH-CL. The experimental dataset, evaluation metrics, parameters setting, ablation experiments, and the comparisons of experimental results are

Algorithm 1 The Training Procedures of CL Phase

Input: Unlabeled data set $\mathcal{X}_u$, number of iterations $epochs$.

Output: The trained MSHFE $f_\theta(\cdot)$.

- 1 Initialize parameters θ of MSHFE $f_\theta(\cdot)$ and nonlinear projector $g_\theta(\cdot)$;
- 2 **for** $e = 1$ **to** $epochs$ **do**
- 3 Obtain mini-batch x by sampling from the unlabeled data set $\mathcal{X}_u$;
- 4 Get two views of x by data augmentation method $Aug(\cdot)$;
- 5 Obtain the latent code h by (1);
- 6 Obtain the projected feature z by (2);
- 7 Calculate contrastive loss $\mathcal{L}_{InfoNCE}$ by (3);
- 8 Update model parameters θ by back-propagating $\mathcal{L}_{InfoNCE}$;
- 9 **end**

presented in detail. All experiments are conducted on a server with an NVIDIA Tesla V100 Graphics Card and PyTorch.

A. Experimental Dataset

In this part, the real-world heavy-plate production process data set is utilized in the experiments. The data examples are collected from a real-world heavy-plate production line located in Eastern China. As shown in Table I, the data set contains the normal plate shape and four kinds of anomaly, including bend, horizon wave, left wave, and right wave.

As the analysis of heavy-plate production process in Section II-B, the 281 shape quality relevant variables are selected. Then, as mentioned in Section III-A, the collected data examples are cleaned and preprocessed first, as shown

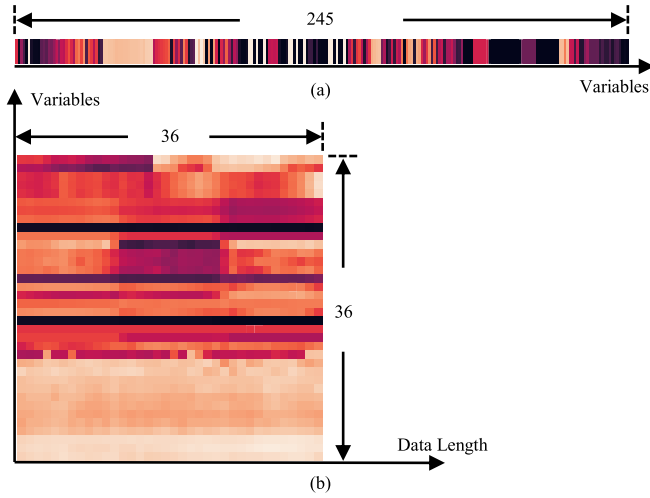


Fig. 10. The schematic diagram of MSH data after cleaning and preprocessing. (a) $x_{1d} \in \mathcal{R}^{1 \times 245}$. (b) $X_{2d} \in \mathcal{R}^{36 \times 36}$.

TABLE I

DATA SET OF HEAVY-PLATE SHAPE COLLECTED FROM THE REAL-WORLD PRODUCTION PROCESS

The description of plate shape quality	Number of samples
Normal	14778
Bend	3209
Horizon wave	2374
Left wave	14397
Right wave	763

in Fig. 10, a data example is turned into two parts, namely a vector $x_{1d} \in \mathcal{R}^{1 \times 245}$ and a matrix $X_{2d} \in \mathcal{R}^{36 \times 36}$.

Finally, the collected data is divided into the CL training set, the fine-tuning set and the testing set according to the ratio of 0.6, 0.2, and 0.2.

B. Evaluation Metric

The evaluation metric, i.e., accuracy, is adopted to illustrate the performance of the proposed approach. Accuracy is the rate of correct classification. The definition of accuracy is formulated as

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}, \quad (12)$$

where TP denotes that both the truth value and predicted value are positive, FP denotes that the truth value is negative but the predicted value is positive, TN denotes that both the truth value and predicted value are negative, FN denotes that the truth value is positive but the predicted value is negative.

C. Parameter Settings

In the training phase of CL, the Adam optimizer is applied with 50 epochs. The batchsize is set to 512. The learning rate and weight decay are set to $1e-4$ and $1e-4$, respectively. The temperature hyper-parameter τ is set to 0.5 as mentioned in [28]. In the fine-tune training phase, the SGD optimizer is applied for 50 epochs. The batchsize is set to 128. The learning rate and weight decay are set to $1e-3$ and $1e-4$, respectively.

TABLE II

THE RESULTS OF MSH-CL WITH DIFFERENT DATA AUGMENTATION METHODS

Methods	α	β	Accuracy
Method A	0.3	-	81.60
	0.1	-	84.48
	0.01	-	83.83
Method B	-	0.3	82.39
	-	0.1	83.23
	-	0.01	83.52
Method C	0.1	0.1	83.81
	0.01	0.01	84.95
	0.01	0.1	85.31
Method D	0.01	0.3	86.04
	0.1	0.1	83.15
	0.01	0.01	85.18
	0.01	0.1	83.48
	0.01	0.3	83.35

The 1D version of ResNet-18 and the naive ResNet-18 is adopted in MSH-FEN as the two pathways backbone to extract the feature of x_{1d} and X_{2d} .

D. Ablation Experiments

In this subsection, the effectiveness of the proposed components in MSH-CL is verified and discussed in detail.

1) *Selection of the Data Augmentation Method*: The data augmentation method is designed to generate two views of the same data sample, which should guarantee that it has similar semantic information. Therefore, four data augmentation methods are designed, and then explored to determine which one is appropriate for this task. The four methods are presented as follows:

- Method A. Adding a certain proportion α of Gaussian noise to the preprocessed data.
- Method B. Performing a certain probability β of dropout on the preprocessed data.
- Method C. Adding a certain proportion of Gaussian noise processed by a certain probability of dropout to the data.
- Method D. Adding a certain proportion of Gaussian noise to the data and then performing a certain probability dropout.

In this ablation investigation, the other components proposed in MSH-CL are set to the same to eliminate the distraction of other factors for the experimental result. The CSMH in MSH-FFN adopts the parallel strategy (See Fig. 8(a)) to fuse the h_{1d} and h_{2d} to obtain the latent code h . The results of different augmentation methods with different α and β values are reported in Table II.

As shown in Table II, the data augmentation has significantly influenced the anomaly detection accuracy on the heavy-plate shape. Method C has achieved the best result with $\alpha = 0.01$ and $\beta = 0.3$. Thus, Method C is adopted as the augmentation method in the subsequent experiments to generate different views of the process data.

2) *Effects of Proposed CSMH Feature Fusion Approach*: The ablation experiment of the MSH data information fusion strategy in MSH-FFN is conducted in this part. As shown in Section III-C2, the proposed CSMH has three various fusion

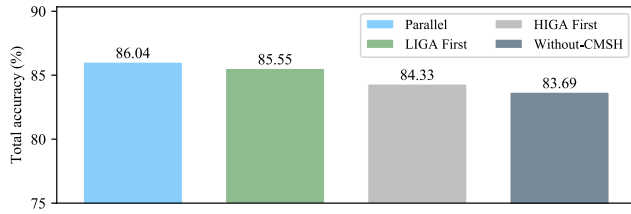


Fig. 11. The results of MSH-CL with different information fusion strategy.

strategies to satisfy the requirement of different application scenarios. In this part, the experiment is first introduced to compare the performance of different CMSH strategies in MSH-FFN. Then, to demonstrate the effectiveness of the proposed fusion approach, it is also compared with the ordinary fusion approach, which concatenates the features of MSH-FEN directly.

As shown in Table III and Fig. 11, the parallel strategy of CMSH has achieved outstanding accuracy in the shape quality anomaly detection task. Meanwhile, the three fusion strategies of CMSH present better performance than the directly concatenating features method. The results verified that the CMSH could improve the capability of the model. Thus, the parallel strategy is adopted in MSH-FFN as the fusion approach for the quality anomaly detection task.

E. Analysis of Experimental Results

This subsection first explores the effects of different fine-tuning set scales on the performance of MSH-CL. Then, compared with other methods, the superiority and effectiveness of MSH-CL are verified and discussed in detail.

1) *Effects of Different Fine-Tuning Set Scales:* In this part, the performance of the proposed MSH-CL with different scale sets utilized in fine-tuning phase is explored. The number of labeled data examples used in this phase may significantly influence the anomaly detection accuracy for heavy-plate shapes. Therefore, the different proportion data examples of the fine-tuning train data set are used to fine-tune the model separately. The results are summarized in Fig. 12. It can be observed that as the number of labeled data examples reduces, the total anomaly detection accuracy decreases at the same time. Also, when the proportion dropped from 10% to 5%, the accuracy decreased sharply because the minority class samples in the fine-tuning phase were rare. However, the MSH-CL can still maintain a higher anomaly detection accuracy with extremely few labeled data for the heavy-plate shape.

2) *Comparisons Between MSH-CL and Other Semi- or Self-Supervised Methods:* In this part, the proposed approach is compared with other semi- or self-supervised methods that are commonly used to solve the less labeled data issue. As mentioned in [16], the co-training strategy with multiple machine learning methods is selected as the baseline models, such as Adaboost, Decision Tree, and support vector machine (SVM). Meanwhile, the Autoencoder and Denoising Autoencoder, which are the typical self-supervised learning methods, will also be included. The preprocessed data is flattened as one-dimensional because these methods cannot directly deal with the MSH data. The results are given in Table IV and

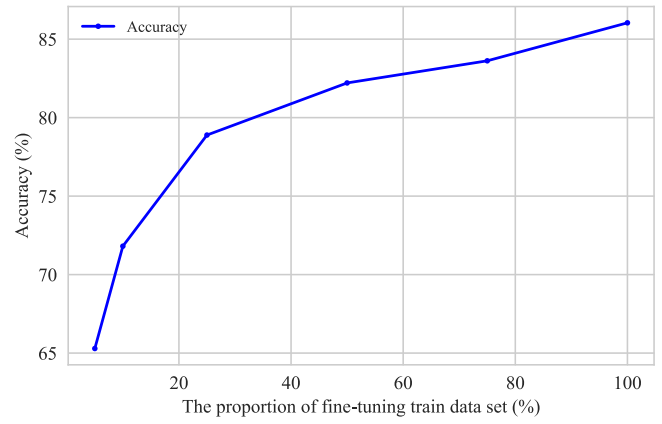


Fig. 12. The results of MSH-CL with the different scale of fine-tuning data set.

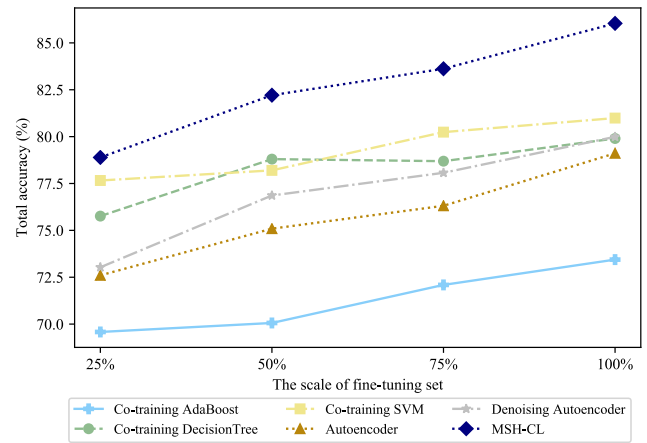


Fig. 13. The accuracy of MSH-CL and other semi- or self-supervised methods with the different scales of fine-tuning set.

Fig. 13. One can see that the proposed MSH-CL can achieve better anomaly detection accuracy than others, and the total accuracy is improved by 12.60%, 6.16%, 5.05%, 6.93% and 6.05% compared with the co-training AdaBoost, co-training Decision Tree, co-training SVM, Autoencoder, and Denoising Autoencoder, respectively. At the same time, it can be seen that the detection accuracy of Autoencoder and Denoising autoencoder for abnormal quality classes with a small number of samples is insufficient. In particular, it is unable to detect the right wave of heavy-plate shape, which is with few samples. For further comparison, the model performance on fine-tuning set with different scales of 25%, 50%, 75% and 100% are also reported in Table IV. It can be seen that the MSH-CL also achieved better results on the labeled data sets with different scales than other methods. In particular, while the proportion of labeled samples reduces, the detection accuracy of the right wave and horizon wave of co-training AdaBoost, co-training SVM, Autoencoder and Denoising autoencoder also decreases.

3) *Comparisons Between MSH-CL and Other Supervised Methods:* In this part, the experiment is conducted where the quality anomaly detection model is trained using the labeled data examples, which demonstrates the merit of CL pre-training with the massive unlabeled data. Thus, the pro-

TABLE III
THE RESULTS OF MSH-CL WITH DIFFERENT FUSION STRATEGY

Strategy	Accuracy					Total Accuracy
	Normal	Bend	Horizon wave	Left wave	Right wave	
Parallel	93.12	65.33	61.98	89.74	36.95	86.04
LIGA First	93.05	60.61	59.31	90.42	34.78	85.55
HIGA First	91.73	64.11	53.38	88.91	32.55	84.33
Without-CMSH	91.71	62.32	51.97	88.04	32.90	83.69

TABLE IV
COMPARISON RESULTS OF MSH-CL WITH OTHER SEMI- OR SELF-SUPERVISED LEARNING METHODS BASED ON HEAVY-PLATE PRODUCTION DATA SET

Methods	Proportion	Accuracy					Total Accuracy
		Normal	Bend	Horizon wave	Left wave	Right wave	
Co-training AdaBoost	100%	84.89	48.98	21.91	78.93	7.33	73.44
	75%	82.56	40.88	39.45	77.62	0.63	72.09
	50%	84.52	13.96	23.65	78.22	3.64	70.06
	25%	86.44	10.23	11.00	77.99	0.66	69.58
Co-training DecisionTree	100%	88.13	55.16	52.39	84.11	25.00	79.90
	75%	87.56	55.13	49.68	82.24	27.78	78.69
	50%	85.94	56.47	55.53	82.41	23.97	78.80
	25%	86.48	51.43	44.37	78.15	22.67	75.76
Co-training SVM	100%	90.85	59.97	15.10	91.23	1.32	80.99
	75%	90.08	58.47	15.93	89.65	0.00	80.24
	50%	89.06	36.80	15.67	91.63	0.00	78.20
	25%	89.28	49.31	8.50	87.77	0.00	77.66
Autoencoder	100%	90.64	41.81	15.13	89.73	0.00	79.11
	75%	91.30	27.69	2.82	88.43	0.00	76.31
	50%	85.94	17.55	5.66	87.96	0.00	75.09
	25%	88.55	4.47	3.47	85.94	0.00	72.60
Denoising Autoencoder	100%	90.97	50.54	16.67	90.13	0.00	79.99
	75%	89.58	46.45	7.38	89.30	0.00	78.07
	50%	89.76	32.64	1.20	89.71	0.00	76.86
	25%	90.48	7.29	2.13	85.47	0.00	73.03
MSH-CL	100%	93.12	65.33	61.98	89.74	36.95	86.04
	75%	91.60	56.71	54.97	89.53	24.81	83.62
	50%	91.51	55.98	41.97	88.31	20.75	82.21
	25%	88.27	42.24	41.06	88.21	7.67	78.89

TABLE V
THE RESULTS OF SUPERVISED LEARNING METHODS BASED ON HEAVY-PLATE PRODUCTION DATA SET

Methods	Accuracy					Total Accuracy
	Normal	Bend	Horizon wave	Left wave	Right wave	
Decision Tree	87.48	59.06	54.49	84.19	44.55	80.41
AdaBoost	83.16	46.17	51.51	76.03	26.92	75.22
SVM	72.12	78.57	64.51	72.62	00.00	72.36
MSH-CL counterpart	93.35	64.68	52.47	85.60	39.33	83.70

posed MSH-CL is compared with a number of supervised algorithms, such as AdaBoost, Decision Tree and SVM. The high-dimensional data also need to be expanded into one-dimensional since these algorithms cannot directly handle the MSH data. The train set of these algorithms is identical to the fine-tuning set used in the second phase of MSH-CL. In this part, the MSH-CL is also compared with its supervised counterpart, which trains the MSHFE in a supervised learning manner. The parameters of the supervised learning counterpart are set the same as in the second phase of MSH-CL. As shown in Table IV, Table V and Fig. 14, the results of MSH-CL are better than its supervised learning counterpart with the same number of the labeled data. Meanwhile, compared with other supervised algorithms, the MSH-CL can also achieve better performance.

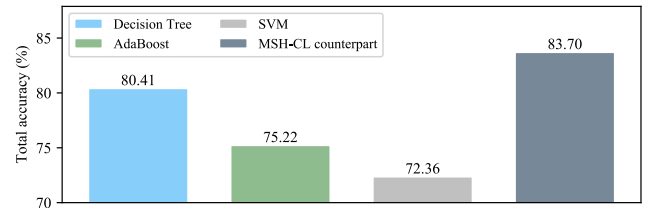


Fig. 14. The accuracy of MSH-CL counterpart and other supervised methods.

V. CONCLUSION

In this paper, the MSH-CL is proposed for quality anomaly detection of the heavy-plate shape, and the CL framework is developed to solve the problem of less labeled data in real-world manufacturing process. Meanwhile, to deal with the

MSH data, a two pathways network to extract data features is designed, and an attention-based feature fusion approach named CMSH is proposed to take advantage of the inner correlation of MSH data. On the one hand, the CL phase can utilize the massive unlabeled data to pretrain the MSHFE to get a feature encoder. On the other hand, the CMSH in MSH-FFN can improve the feature fusion performance by utilizing the inner correlation. The ablation experiments and comparative experiments are conducted on the dataset which collected from the real-world heavy-plate production process. The experimental results indicate that the MSH-CL can improve the performance of plate shape anomaly detection with the less labeled and MSH data.

In future works, the proposed approach will be extended to other quality-relevant detection problems present in the heavy-plate production process. Meanwhile, more practical factors, such as different kinds of steel slabs, and some limitations of the proposed approach will also be considered.

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